Q(V, ϕ) parametrization — method

ETL LGAD charge vs bias and fluence

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This note describes, step by step, the laws used to parametrize the charge Q collected by ETL LGADs as a function of the bias V and the neutron–equivalent fluence ϕ. It documents the code in this repository (`data_io.py`, `model.py`, `plots.py`, `config.py`).

1 Data and pre-processing

Measurements live in `parametrization_data.xlsx`, one *sheet per production* (e.g. `HPK`, `FBK-LF`). Inside a sheet each sensor scan is a block of three columns V, Q, Res; the block label carries the sensor name and the fluence (e.g. `HPK6 S4-B - 1e15`). The reader extracts:

- **production** = sheet name;
- **fluence** parsed from the label (0 if none is written);
- **setup** = `TB` if the label mentions a test beam, else `Beta`.

Pooling. For a given production and fluence, the (V, Q) points of all `Beta` sensors are *pooled* into a single data set and fitted together; test–beam (`TB`) points are shown only as a cross–check and are never fitted. Within one production sensors of nominally equivalent generations may be pooled across fluence (e.g. HPK6 and HPK2).

Uncertainties. The raw data has no error bars, so constant values are assigned (editable in `config.py`):

$$\sigma_Q = 1\,\text{fC (switchable to 2 fC)}, \qquad \sigma_V = 2\,\text{V}, \qquad \sigma_\phi = 0.10\,\phi. \tag{1}$$

A constant σ_Q does not bias the central fit values; it only sets the weighting and the reported parameter errors.

2 Step 1 — charge vs bias at fixed fluence

At each fluence the pooled $Q(V)$ points are fitted with

$$\boxed{Q(V) = p_0 + p_1\,V + \exp\big(p_2 + p_3\,V\big)} \tag{2}$$

a linear term plus an exponential (gain–like) rise.

Analytic seeding (no manual tuning). ROOT's default start values make this fit unstable, which previously required hand–tuned parameter limits. Instead the seed is computed from the data (`auto_seed_QV`):

1. sort by V; fit a straight line to the lowest ∼40% of the points to get the baseline (p_0, p_1);
2. form the residual $R(V) = Q - (p_0 + p_1 V)$ and, on the points with $R > 0$, fit $\ln R = p_2 + p_3 V$ to get (p_2, p_3).

With this seed the fit converges on its own.

Physical bounds. The only constraints are *uniform* (the same for every fit, not per point):

$$p_1 \geq 0, \qquad p_3 \geq 0, \tag{3}$$

i.e. the collected charge does not decrease with bias. The fit is performed in ROOT (`TGraphErrors::Fit` of Eq. 2); if ROOT fails to return a valid result, a `scipy.optimize.curve_fit` with the same seed is used as a silent fallback.

The outputs are the four parameters $p_k(\phi_j)$ and their errors at each measured fluence ϕ_j, with the fit χ^2/ndf.

3 Step 2 — parameters vs fluence

Each parameter p_k is then modelled as a function of ϕ. Four laws are available (`config.PARAM_MODE`); let $\{(\phi_j, p_{k,j}, \sigma_{k,j})\}$ be the points from Step 1.

(i) Node–exact interpolation `interp` (default). Piecewise–linear through the measured points, with linear extrapolation beyond the end nodes:

$$p_k(\phi) = p_{k,j} + \frac{p_{k,j+1} - p_{k,j}}{\phi_{j+1} - \phi_j}\,(\phi - \phi_j), \qquad \phi \in [\phi_j, \phi_{j+1}]. \tag{4}$$

By construction it reproduces every measured fluence exactly, so the predicted $Q(V, \phi)$ matches the Step–1 fit at each ϕ_j.

(ii) Linear `lin`. Weighted least squares,

$$p_k(\phi) = m_k\,\phi + q_k, \qquad \text{weights } w_j = 1/\sigma_{k,j}. \tag{5}$$

(iii) Exponential `exp`.

$$p_k(\phi) = a_k + \exp\big(b_k\,\phi + c_k\big), \tag{6}$$

fitted in ROOT with an analytic seed (a_k near the asymptote, (b_k, c_k) from a log fit of $p_k - a_k$).

(iv) Automatic `auto`. If there are fewer than `MIN_PTS_FOR_EXP` $= 4$ fluences, use `lin`. Otherwise fit both `lin` and `exp` and keep the one with the smaller Akaike information criterion,

$$\mathrm{AIC} = \chi^2 + 2k, \tag{7}$$

with k the number of free parameters (2 for `lin`, 3 for `exp`). This is decided independently for each p_k.

Why `interp` is the default here. The form of Eq. 2 is partly degenerate: p_0, p_1, p_2 are strongly anti–correlated, so while the *curve* $Q(V)$ is well constrained the *individual* parameters are not and scatter from fluence to fluence. As a result the measured $p_k(\phi)$ do not follow a smooth low–order law, and a global `lin`/`exp` fit of p_2, p_3 — to which Q is exponentially sensitive — can mispredict the charge badly. The node–exact `interp` avoids this by reproducing the data at every measured fluence; `auto`/`exp` remain available when a smooth, extrapolable surface is preferred.

4 Combined surface and its inverse

Substituting the parameter laws into Eq. 2 gives the two–dimensional surface

$$\boxed{Q(V, \phi) = P_0(\phi) + P_1(\phi)\,V + \exp\big(P_2(\phi) + P_3(\phi)\,V\big)} \tag{8}$$

with $P_k(\phi)$ one of the laws above. When every P_k is `lin` or `exp` this is exported as a single closed–form ROOT `TF2` (`build_QVphi_TF2`); for `interp` the surface is evaluated numerically (`Q_of_V_phi`).

Bias for a target charge. To get the bias needed to reach a charge $Q^\star$ at a fluence ϕ, the equation $Q(V,\phi) = Q^\star$ is inverted numerically (`V_for_Q`): $Q(V,\phi)$ is scanned on a dense V grid and the *highest* upward crossing of $Q^\star$ is returned (robust to any non–monotonic behaviour of a degenerate fit); `NaN` if $Q^\star$ is never reached.

5 The global simultaneous model

Fitting each fluence *independently* (Sec. 2) and then fitting the resulting parameters is unreliable here: at a fixed fluence the four parameters are strongly degenerate (the exponential can mimic the polynomial over the measured V range), so the per–fluence values do *not* follow any smooth trend — e.g. the pure–gain exponent p_2 jumps back up at the highest fluence instead of continuing to fall. Independent laws then inherit this scatter and the surface is poor at high fluence.

The cure is to impose the smoothness *during* the fit: a single *global simultaneous* least squares to all (V,ϕ,Q) points at once, in which each voltage parameter is replaced by its law of fluence,

$$\boxed{Q(V,\phi) = p_0(\phi) + p_1(\phi)\,V + \exp\big(p_2(\phi) + p_3(\phi)\,V\big)} \tag{9}$$

$$p_k(\phi) = a_k + \exp\big(b_k\,\phi + c_k\big) \quad (k = 0,1,3), \qquad p_2(\phi) = q_2 + m_2\,\phi. \tag{10}$$

The gain exponent p_2 is taken *linear* on purpose: the gain amplitude is then $\exp(p_2) = \exp(q_2)\,e^{m_2\phi}$, a *single* exponential decay in fluence (acceptor removal), gradual just above $\phi = 0$ and vanishing at high ϕ (with $m_2 < 0$). An exponential law for p_2 would make $\exp(p_2)$ a *double* exponential — far too steep right after $\phi = 0$, which over–predicts the bias needed at low fluence (e.g. it wrongly asked for ∼290 V instead of ∼200 V to reach 15 fC at $\phi = 1\times10^{14}\,\mathrm{n_{eq}/cm^2}$). fitted together with `scipy.optimize.curve_fit`. Because the fluence dependence is forced through the smooth laws of Eq. 10, the degeneracy at each fluence is resolved by the neighbouring fluences — among the near–equivalent per–fluence parameter combinations the fit selects the one lying on a smooth trend. The high–fluence p_2 outlier disappears and, for HPK, the fit reaches $\chi^2/\mathrm{ndf} \approx 28/48$ and a prediction RMS of 0.66 fC on the real points. By default p_0 carries its own law (Eq. 10, $k = 0$); `config.P0_VS_PHI` can instead make it a single global constant. Each parameter law is printed with its fitted numbers on the parameter–law figure.

Physical constraints (no per–point tuning). Only uniform constraints are applied:

$$a_1, a_3 \geq 0 \;\; (p_1, p_3 \geq 0:\ \text{charge rises with bias}), \qquad p_0(\phi) \in [\,0,\,1\,]\ \text{fC} \;\; (\texttt{config.P0_BOUNDS}). \tag{11}$$

The last line bounds the *high–fluence asymptote*: once the gain is gone the collected charge is the bulk (no–gain MIP) charge, a small number between 0 and ∼ 1 fC. p_0 is seeded at the per–production bulk value Q_∞ (`config.Q_INF`: HPK 0.50 fC, FBK 0.55 fC).

High–fluence convergence. At very high fluence the gain layer is fully removed, so for *every* production $Q \to Q_\infty$, flat in V. For HPK (five fluences, with low–charge points at $\phi = 1.5\times10^{15}$) this happens on its own. For a sparsely–measured production it cannot: FBK is measured only up to $1.5\times10^{15}\,\mathrm{n_{eq}/cm^2}$, where Q is still 2–10 fC (the gain is far from gone), so the *rate* at which the gain dies beyond the data is simply not measured, and the bare extrapolation plateaus well above Q_∞. Parameter bounds alone cannot fix this without either leaving the plateau or distorting the in–data fit. We therefore add a *soft physical prior* on the known asymptote (`config.ANCHOR_FLOOR`): a few points $Q = Q_\infty$ (flat in V) at high fluences `ANCHOR_PHIS`, weighted by `ANCHOR_ERR`. It is a prior on the asymptote, not a hard clamp: it acts only far beyond the data, the reported χ^2/RMS use the *real* points only, and the in–data fit is unchanged (HPK/FBK RMS 0.66/0.64 fC). The surface then decays smoothly to Q_∞ (≈ 0.5–0.6 fC), matching the simulation.

Uncertainties and inverse. The parameter covariance is propagated to a band on each $p_k(\phi)$ (shown on the parameter–law figure). The bias for a target charge $Q^\star$ is obtained by inverting Eq. 9 numerically (`V_for_Q`): $Q(V,\phi)$ is monotonic in V (since $p_1, p_3 \geq 0$), so the single crossing of $Q^\star$ is returned, or `NaN` if it is never reached. This model is `model.SimulSurface` / `model.fit_global`; the per–fluence fits of Sec. 2 remain as diagnostics.

6 Operating point with the SEB limit

Each production has a single–event–burnout ceiling V_{SEB} (`config.V_SEB`; HPK 550 V, FBK 605 V) above which it cannot be operated. For a target charge $Q^\star$ (default `PLATEAU_CHARGE` = 15 fC):

$$\text{bias needed:} \quad V(\phi) = \min\big(V_{\mathrm{SEB}},\, V_{\mathrm{for}\,Q^\star}(\phi)\big), \tag{12}$$

$$\text{charge obtainable:} \quad Q_{\mathrm{obt}}(\phi) = \min\big(Q^\star,\, Q(V_{\mathrm{SEB}},\phi)\big). \tag{13}$$

Below the "SEB knee" the target is reached within V_{SEB} so $Q_{\mathrm{obt}} = Q^\star$ (flat); beyond it the device is stuck at V_{SEB} and delivers $Q(V_{\mathrm{SEB}},\phi)$, which falls with fluence towards Q_∞. These are the solid (charge, left axis) and dashed (bias, right axis) curves of the operating–point figure, drawn once per production.

7 Comparison with the simulation

For a cross–check the operating–point figure can overlay a simulation curve read from `simulation/<production>` (columns fluence, charge, bias). The data–driven parametrization (azure for HPK, red for FBK) and the simulation (green for HPK, gold for FBK) are drawn on the same axes: the solid lines are the obtainable charge, the dashed lines the bias needed, both capped at V_{SEB}. The two agree on the SEB knee position and on the high–fluence plateau.

8 Caveats

- The high–fluence asymptote of a sparsely–measured production (e.g. FBK, three fluences) is *not* determined by its data: it is set by the bulk prior Q_∞ and the bound $p_0 \in [0,1]$. The curve there is a constrained extrapolation, not a measurement — adding fluence points above 1.5×10^{15} would replace the prior with data.
- Extrapolation in V beyond the measured range, and in ϕ beyond the last measured fluence, is model dependent and should be treated with care.
- The per–fluence parameters are partly degenerate; prefer the *curves*/predictions over the individual Step–1 parameter values.